

4. Cancers are caused by disruption of the normal cell cycle and can involve proto-oncogenes and epigenetic factors. Some cancers can be treated using chemotherapy which also kills some of the non-cancerous body cells. After treatment the tissues can be repopulated using stem cells.

(a) Define what is meant by the following terms:

[3]

(i) Proto-oncogene

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(ii) Epigenetics

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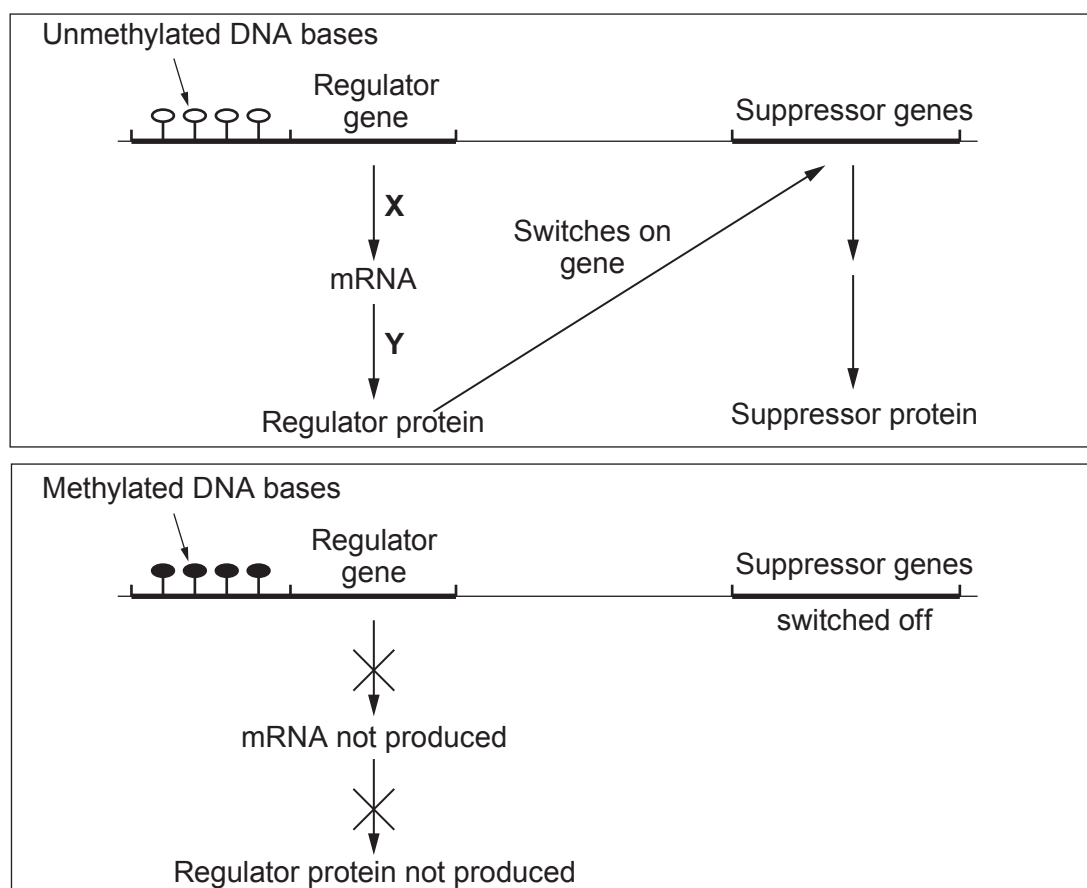
(iii) Stem cell

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- (b) In the human genome there are suppressor genes which regulate cell division and growth. These genes can be switched off (silenced) to increase cell division or growth, and switched on to suppress division and growth.

Two molecular processes are vital in the silencing of genes, DNA methylation and histone modification.

The diagram below shows the effect of DNA methylation.



- (i) Name the processes labelled **X** and **Y** on the diagram. [1]

X

Y

- (ii) Cancer cells can cause DNA methylation in non-cancerous cells. With reference to the diagram explain how DNA methylation results in tumour growth. [3]

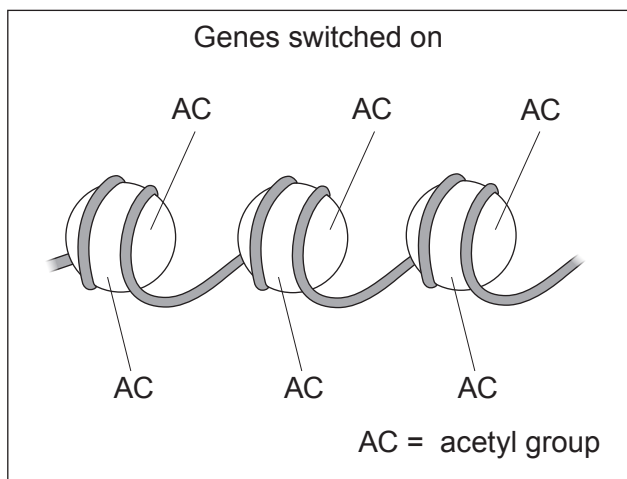
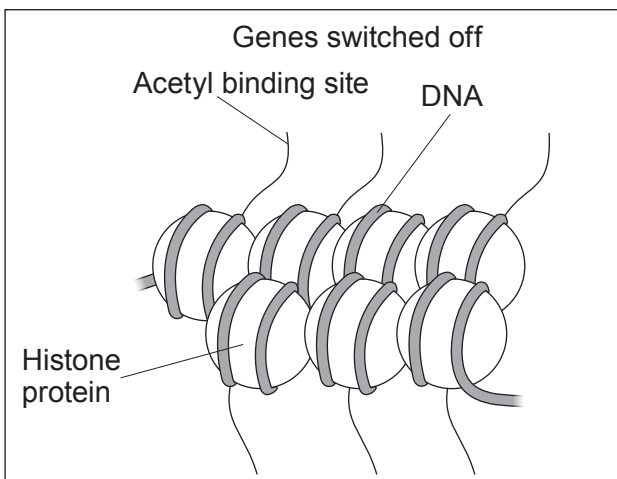
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Histones are proteins which have DNA wrapped around them as shown in the diagram below. The histones can be acetylated by having an acetyl group added to them by an enzyme. Acetylated histones have an open structure which allows RNA polymerase to have access to the DNA and the genes are switched on.



- (iii) In relation to the diagram, suggest how cancer cells switch the suppressor genes off. [3]

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- (iv) In some humans, tumour suppressor genes can become permanently switched off in **all cells** in the body. Explain the consequences of such a change. [2]

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